Econometrics I

Lecture 2b: Transformations — Logs, Odds, and Interpretation

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Why transform anything?

Three distinct reasons, often confused:

1. Interpretation: you want the coefficient to answer the question people actually ask — “what’s the *percentage* effect?”, “what’s the *elasticity*?”
2. Functional form: the true relationship is multiplicative or has diminishing returns, and a transformation makes it linear-in-parameters.
3. Statistical behavior: skewed outcomes, variance that grows with the level, outliers with huge leverage.

Most business data (revenue, prices, firm sizes, salaries, clicks) checks all three boxes — which is why “economists love logs.”

The four regressions

Model	Specification	β_1 means	Canonical business use
level–level	$y = \beta_0 + \beta_1 x$	Δy per unit Δx	test scores on class size
log–level	$\ln y = \beta_0 + \beta_1 x$	$100\beta_1$ % Δy per unit Δx	salary on years of tenure
level–log	$y = \beta_0 + \beta_1 \ln x$	$\beta_1/100$ Δy per 1% Δx	sales on ad spend
log–log	$\ln y = \beta_0 + \beta_1 \ln x$	elasticity: % Δy per % Δx	demand on price

- The log–log slope is the price elasticity of demand when y is quantity and x is price: $\beta_1 = -2$ means a 1% price increase loses 2% of units.
- Bonus: log–log coefficients don't depend on units (dollars vs. euros, ounces vs. liters) — you verified this in Lecture 2.

Percentage thinking: why growth is multiplicative

- Business quantities compound: revenue grows 10% then 10% again, not \$10M then \$10M again. Asset returns, user growth, prices under inflation — all multiplicative.
- Logs turn multiplicative into additive:

$$y_t = y_0(1+g_1)(1+g_2)\cdots \quad \Longrightarrow \quad \ln y_t = \ln y_0 + \textstyle\sum_t \ln(1+g_t)$$

- A process of many small multiplicative shocks yields a lognormal outcome (Gibrat's law) — one story for why firm sizes, incomes, and city sizes look lognormal-ish.
- So a linear model in $\ln y$ is the natural home for anything that grows by percentages.

Log points vs. percent: when the approximation breaks

- In a log–level model, the exact effect of a one-unit change in x (or a dummy switching on) is

$$100\left(e^{\beta_1} - 1\right)\% \qquad \text{not} \qquad 100\,\beta_1\%$$

- Fine for small coefficients: $\beta = 0.05$ is 5.1%. Not fine for big ones:

β	"100β%"	exact
0.05	5%	5.1%
0.30	30%	35%
0.70	70%	101%
-0.70	-70%	-50%

- Note the asymmetry: $+0.7$ and -0.7 log points are $+101\%$ and -50% — which is also why "up 50% then down 50%" loses money.

Skewness: the statistical case for logs

- Firm revenue, CEO pay, household income, order sizes, house prices: heavily right-skewed. A few Walmarts and many corner stores.
- In levels, the biggest observations dominate everything: they drive the mean, the fit, and (via leverage, Lecture 3) your coefficients.
- In logs, Walmart is a point like any other. Residuals look closer to normal, and no single firm decides your estimate.
- Related: variance usually grows with the level — a $10B firm's sales fluctuate by more *dollars* than a $1M firm's, but often by a similar *percentage*. Logs stabilize the variance (less heteroskedasticity, same robust SEs anyway).
- Accounting-journal cousin: winsorizing at the 1st/99th percentile — same motivation (taming extreme observations), different tool, and it changes the estimand too.

Log x only: diminishing returns

- $y = \beta_0 + \beta_1 \ln x$: each doubling of x adds the same amount ($\beta_1 \ln 2$) to y.
- The natural shape for inputs with diminishing returns:
  - Sales on advertising spend: the tenth million of ad dollars buys less than the first.
  - Output on firm size; utility on income; response rates on email volume.
- A famous log–log version: CEO pay on firm size has an elasticity around $1/3$ across firms, decades, and countries — try that one in levels and you learn nothing.
- Practical reading: “moving a store from the 10th to the 90th percentile of foot traffic” is often more honest than “+1 visitor.”

Dummies with log y: premia

When y is logged, coefficients on dummies are percentage premia:

- Union premium, gender gap (Lecture 4's workshop), CEO–MBA premium.
- Marketing versions: the brand premium (branded vs. generic price gap), the verified-badge premium on a platform, organic vs. conventional.
- Accounting versions: the Big-4 audit fee premium; the loan-spread discount for firms with cleaner disclosures.

Remember the previous slide: report $100(e^{\beta} - 1)\%$ when β is large, and remember these are conditional premia — conditional on whatever else is in the regression.

The log(0) problem

- Sales this week, patents this year, exports to a country: often exactly zero, and $\ln 0 = -\infty$.
- Old fixes: $\ln(1 + y)$, or the inverse hyperbolic sine $\text{arcsinh}(y)$. Both look log-like for large y and are defined at 0.
- The modern warning (Chen and Roth, QJE 2024): with zeros, a “percentage effect” is not well-defined, and estimates from log-like transformations depend on the units of y (dollars vs. cents changes your treatment effect!). That’s not a bug in the software; the estimand itself is incoherent.
- Better answers:
  - Poisson regression on the level of y — coefficients are still (semi-)elasticities, zeros are fine. (Session 13; this is why trade economists all switched.)
  - Or split the question: extensive margin (any sales?) and intensive margin (log sales given sales > 0).

Bounded outcomes: odds and log-odds

- Conversion rates, churn rates, default rates live in $[0,1]$ — a linear model happily predicts -4% or 130%.
- Transform in two steps:

$$p \longrightarrow \underbrace{\frac{p}{1-p}}_{\text{odds } \in (0,\infty)} \longrightarrow \underbrace{\ln \frac{p}{1-p}}_{\text{log-odds } \in (-\infty,\infty)}$$

- A 2% conversion rate is odds of 0.0204, log-odds of -3.89. Log-odds are unbounded and (roughly) the right scale for effects to be additive.
- Modeling log-odds as $x'\beta$ is logistic regression — coming in Session 12. Coefficients are changes in log-odds; e^{β} is an odds ratio.

Rare events: relative vs. absolute

- When p is small, odds $\approx p$, so a change in log-odds $\approx$ a percentage change in the probability itself. A logit coefficient of 0.69 on a rare outcome $\approx$ doubling the probability.
- This is where headlines mislead: “doubles the risk of churn” might mean $0.1\% \rightarrow 0.2\%$. Relative effects on rare events can be huge while absolute effects are trivial (and vice versa: a 1pp lift on a 60% base rate is a modest 1.7% relative change and an enormous amount of money).
- Discipline: for any rare-event result, report both the relative effect and the baseline. “Fraud risk doubles, from 2 in 10,000 to 4 in 10,000.”
- Same discipline for your own regressions: a big elasticity on a tiny base is not a big business.

Other transformations you will meet

- Standardizing (z-scores): coefficients in standard-deviation units. Common in strategy/OB when units are survey scales with no natural meaning ("a 1 SD increase in management quality raises productivity by 0.2 SD" — Bloom–Van Reenen).
- Winsorizing / trimming: standard in accounting and finance to control outliers in ratios (ROA, leverage). Know that it changes the estimand, and report the cutoffs.
- Ranks / percentiles: robust to outliers, throws away magnitude ("moving from the 25th to 75th percentile of…").
- Polynomials and splines: transformations of x that stay linear-in-parameters — the $exper^2$ term you have already met.

Checklist: to log or not to log

Log y when:

- it is positive and right-skewed, grows/compounds in percentages, and the question is naturally "what % effect?"

Think twice when:

- y has zeros (use Poisson or split margins — not ln$(1 + y)$);
- the decision-relevant quantity is in levels (profit in dollars — a CFO cannot spend log dollars);
- y is a share or rate in $[0, 1]$ (think odds/logit instead).

Always:

- say in one sentence what your coefficient means, in units a manager understands — if you can't, the transformation chose you, not the other way around.

Thanks!